Kanban.PDF

Mudit Chaturvedi_PES1UG24CS279-BPS#33

SRN: PES1UG24CS279 | Name: Mudit Chaturvedi

Problem Statement #33 — Micro-Lending & Peer Credit Risk Assessor

Jira site: mudit300477.atlassian.net

Project: **Kanban_BPS#33** | Key: **KBPS33** (company-managed Kanban)

Template: Software Development — Kanban

1. Project setup

A company-managed Kanban project was created for Lab 1 scenario #33. The project name is Kanban_BPS#33 and the project key is KBPS33 (Jira keys cannot contain '#'). Work is visualised on the KBPS33 board with columns Backlog → Selected for Development → In Progress → Done.

2. Mapping Lab 1 requirements to the board

All 5 functional requirements and 2 non-functional requirements from Lab 1 were added as board work items. Each requirement is represented as one Epic (scenario), one User Story (use case), one Task, and one Sub-task.

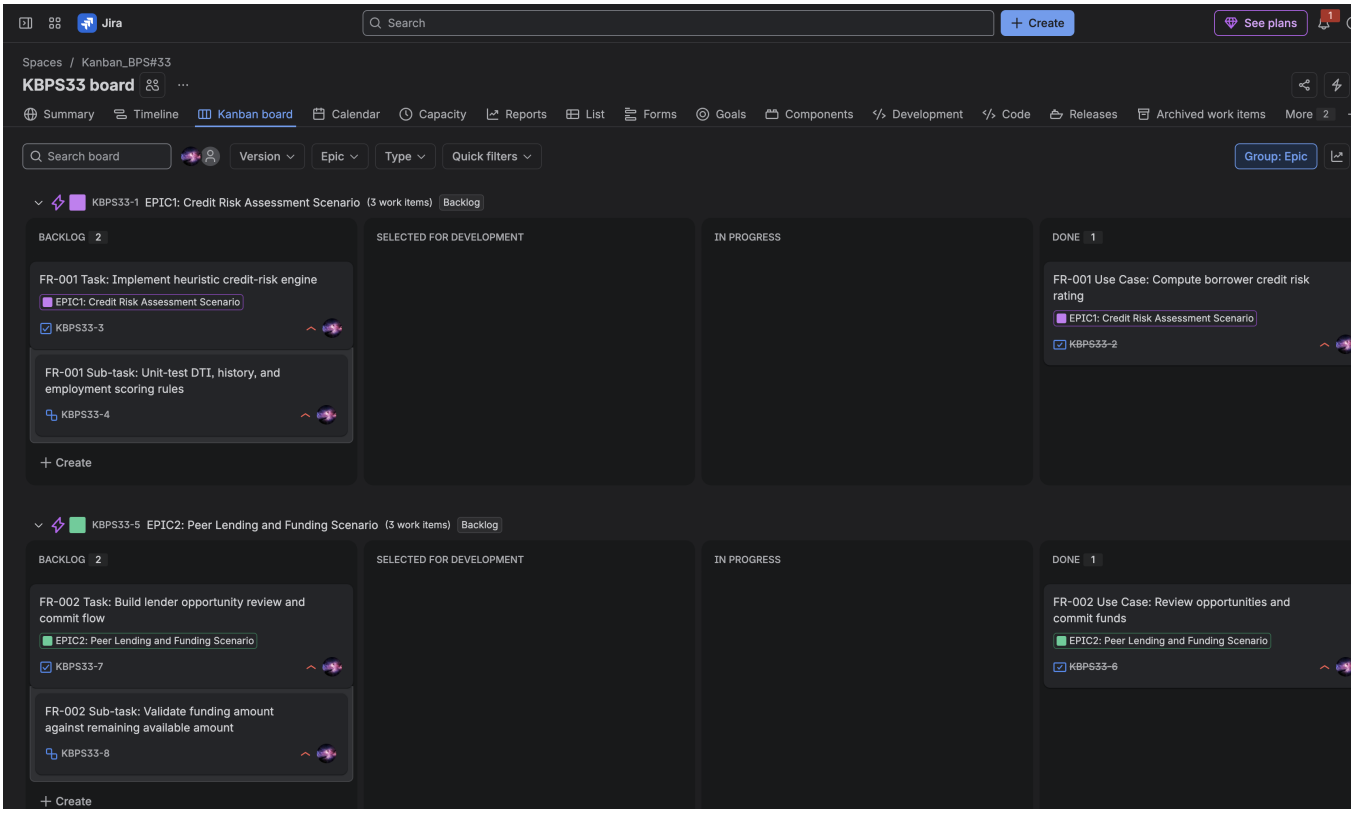


Figure. KBPS33 Kanban board grouped by Epic, showing FR-001 and FR-002 stories, tasks and sub-tasks moving across columns.

3. Epics (scenarios) assigned to board work

Seven epics capture the Micro-Lending & Peer Credit Risk Assessor scenarios:

- EPIC1 KBPS33-1 — Credit Risk Assessment Scenario (FR-001)
- EPIC2 KBPS33-5 — Peer Lending and Funding Scenario (FR-002)
- EPIC3 KBPS33-9 — Repayment Schedule Scenario (FR-003)
- EPIC4 KBPS33-13 — Loan Repayment Recording Scenario (FR-004)
- EPIC5 KBPS33-17 — Default Warning Scenario (FR-005)
- EPIC6 KBPS33-21 — Transaction Integrity Scenario (NFR-001)
- EPIC7 KBPS33-25 — Data Protection Scenario (NFR-002)

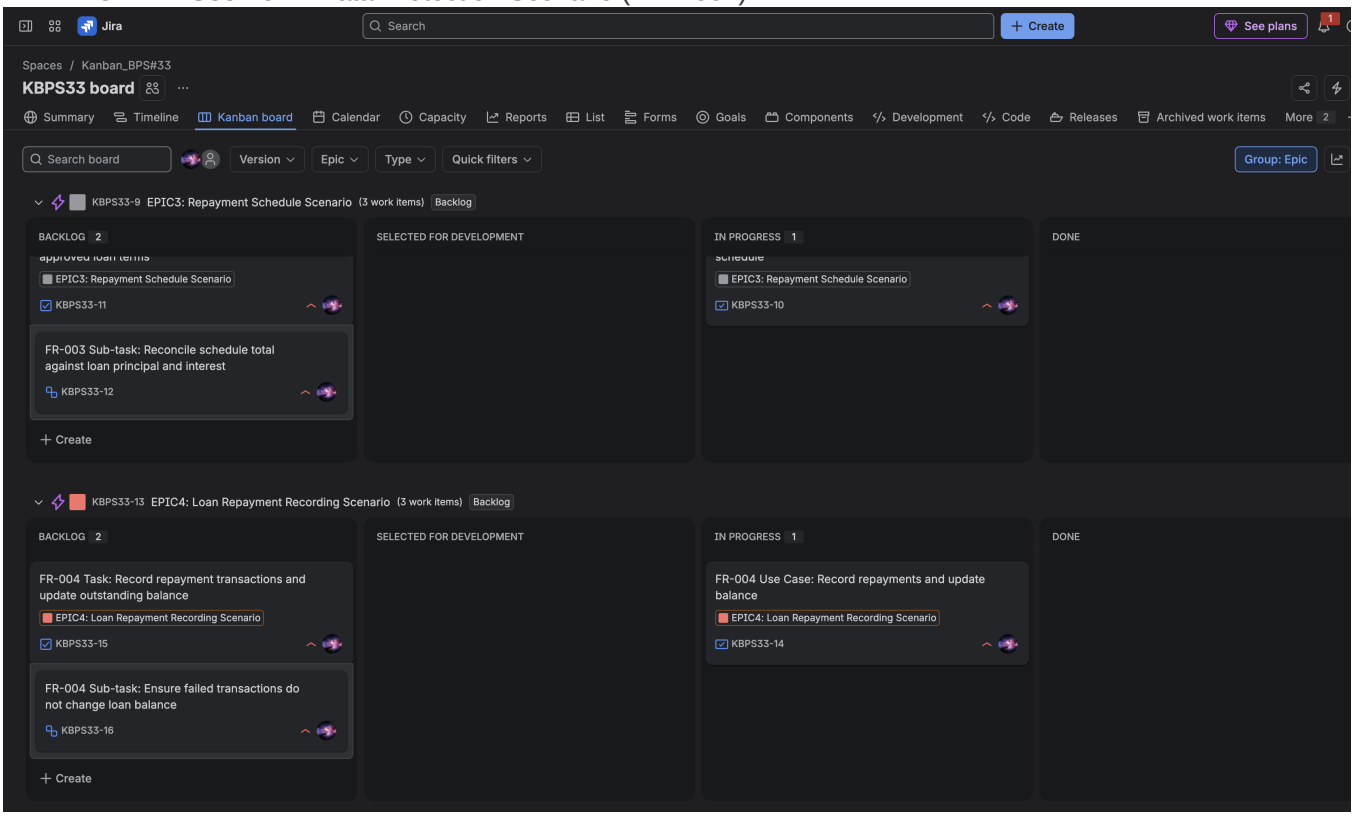


Figure. List view of Kanban_BPS#33: all seven epics with child use-case stories and implementation tasks, assigned to Mudit Chaturvedi.

4. User stories for all 7 requirements (use cases)

- FR-001 Story KBPS33-2 (8 pts): As a peer lender, I want a Low/Medium/High risk tier from DTI, repayment history and employment stability, so that I fund only eligible borrowers.
- FR-002 Story KBPS33-6 (5 pts): As a peer lender, I want to review eligible loans and commit a valid amount, so that I can participate in funding.
- FR-003 Story KBPS33-10 (5 pts): As a borrower, I want a structured installment schedule, so that I know amounts, due dates and remaining balance.
- FR-004 Story KBPS33-14 (5 pts): As a borrower, I want successful repayments to reduce outstanding balance, so that my loan status stays accurate.
- FR-005 Story KBPS33-18 (5 pts): As a borrower, I want an automated warning only when a repayment is overdue, so that I can act before default.
- NFR-001 Story KBPS33-22 (8 pts): As a platform operator, I want repayment and EMI distribution inside ACID transactions, so that funds cannot double-spend.
- NFR-002 Story KBPS33-26 (8 pts): As a borrower, I want authenticated access and encryption in transit/at rest, so that my financial PII is not disclosed.

5. Tasks and sub-tasks under every FR/NFR

Each story has an implementation Task and a verification Sub-task (for example, FR-001 Task: heuristic credit-risk engine; FR-001 Sub-task: unit-test DTI/history/employment rules). Tasks and sub-tasks appear on the same Kanban board under their parent epic.

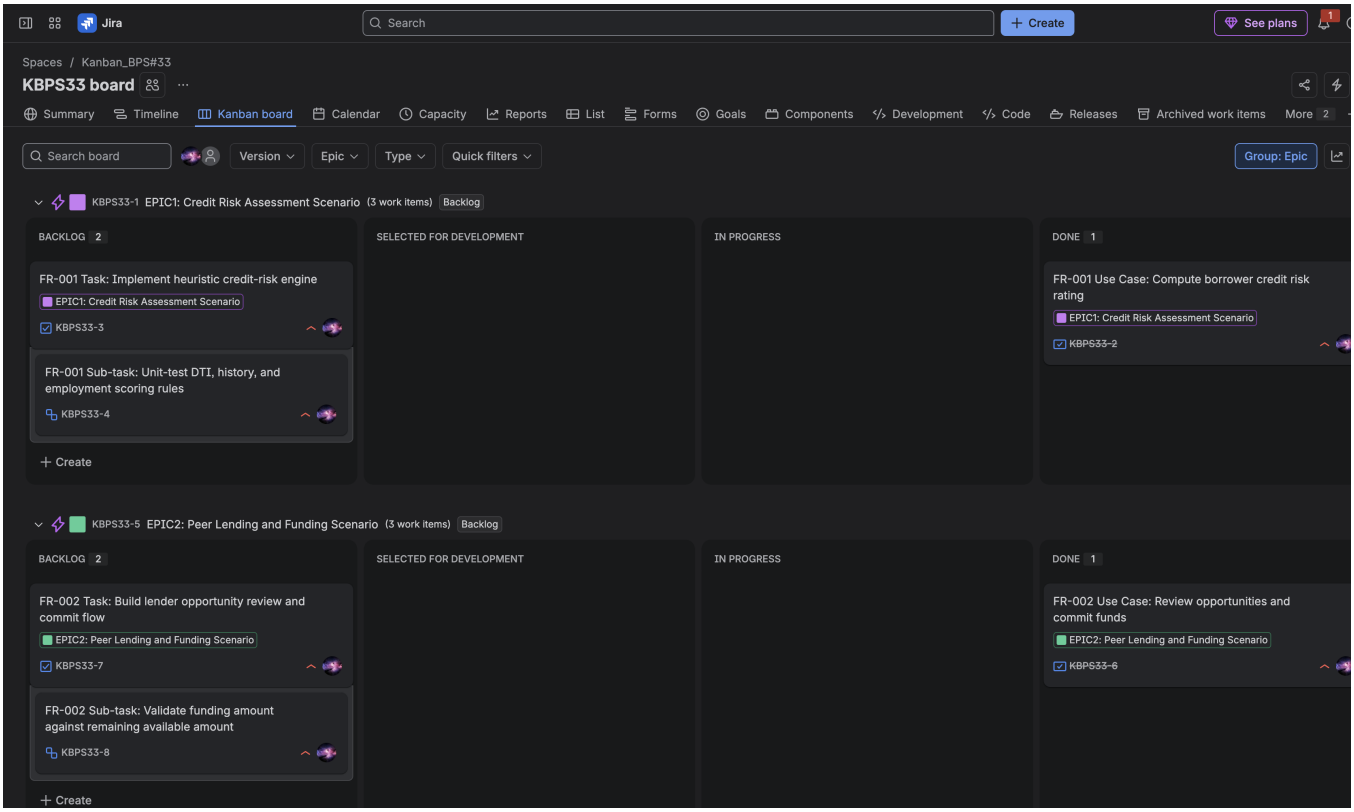


Figure. Kanban board continued: additional epic swimlanes with FR tasks/sub-tasks in Backlog and In Progress, and completed use-case stories in Done.

6. Backlog with every item linked to an Epic

The backlog/list is ordered by epic. Every story, task and sub-task is parented to its scenario epic so the instructor can filter by FR-001…FR-005 and NFR-001/NFR-002. Priority is High for core lending FRs and Highest for NFR-001 (financial integrity).

Sequence of screenshots

1) Board creation / FR+NFR cards on the Kanban board. 2) Epics assigned to board work. 3) Hierarchical list of stories, tasks and sub-tasks. 4) Remaining epic swimlanes on the board.